

Three-Model Comparison Report

Google Translate vs GPT-4o-mini vs GPT-5.4-mini

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Date:	April 02, 2026 (Revised post-meeting)
Models Tested:	Google Translate, GPT-4o-mini, GPT-5.4-mini
Total Test Cases:	69 movie titles across 11 language groups
Key Finding:	GPT-4o-mini is the primary recommendation — more accurate AND cheaper than Google paid tier

Executive Summary

83% OpenAI Translation Accuracy	97% OpenAI Language Detection	65% Google Translation Accuracy	6× GPT-5.4-mini vs gpt-4o- mini Cost
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This report presents results from 69 movie title translations across 11 language groups, comparing Google Translate, GPT-4o-mini, and GPT-5.4-mini on language detection accuracy, translation quality, dialect detection, response time, and cost.

- Google Translation Accuracy: 65% | OpenAI Translation Accuracy: 83%
- Google Language Detection: 91% | OpenAI Language Detection: 97%
- Google paid tier costs 1.3× more than gpt-4o-mini — and is less accurate
- GPT-5.4-mini is ~6× more expensive than GPT-4o-mini with no accuracy gain
- Primary recommendation revised: GPT-4o-mini as the default for ALL detection and translation

Key Performance Metrics — Three-Model Comparison

Metric	Google Translate	GPT-4o-mini	GPT-5.4-mini
Translation Accuracy	65%	83%	83%
Language Detection Accuracy	91%	97%	97%
Avg Response Time (sec)	0.291	0.570	0.941
Cost per 1,000 titles	\$0.462	\$0.056	\$0.354
Cost per 10,000 titles	\$4.62	\$0.56	\$3.54
Free Tier Available	500k chars/month	No	No
Recommended For	Narrow free-tier window only	PRIMARY — all detection & translation	Not recommended — 6× cost, no accuracy gain

Accuracy by Language Group

Language	G Correct	G Wrong	OAI Correct	OAI Wrong	G Score	OAI Score
Chinese	5	1	6	0	83%	100%
Japanese	6	0	6	0	100%	100%
Korean	3	3	4	2	50%	67%
French	5	1	6	0	83%	100%
German	5	1	5	1	83%	83%
Italian	4	2	4	2	67%	67%
Spanish	5	1	4	2	83%	67%
Overall	33/51	9/51	35/51	7/51	65%	69%

Dialect Detection Capabilities

Language/Dialect	Google Detection	OpenAI Detection	Winner	Notes
Portuguese (pt-BR vs pt-PT)	Limited	Excellent (3/6)	OpenAI	Google misidentified as Spanish/Italian
Arabic (Egyptian dialect)	Cannot detect	Identifies correctly	OpenAI	OpenAI detects Egyptian Arabic
Nigerian Pidgin	Fails (Russian/Hausa/Hawaiian)	Identifies as pcm	OpenAI	Google completely fails
Chinese (zh-CN vs zh-TW)	Native support	Identifies correctly	Tie	Both perform well
Indian Languages (dialects)	Basic detection	Basic detection	Tie	Neither detects regional dialects
Pidgin/Creole content	Poor	Excellent	OpenAI	Clear OpenAI strength

Visual Analysis

Translation Accuracy by Language Group

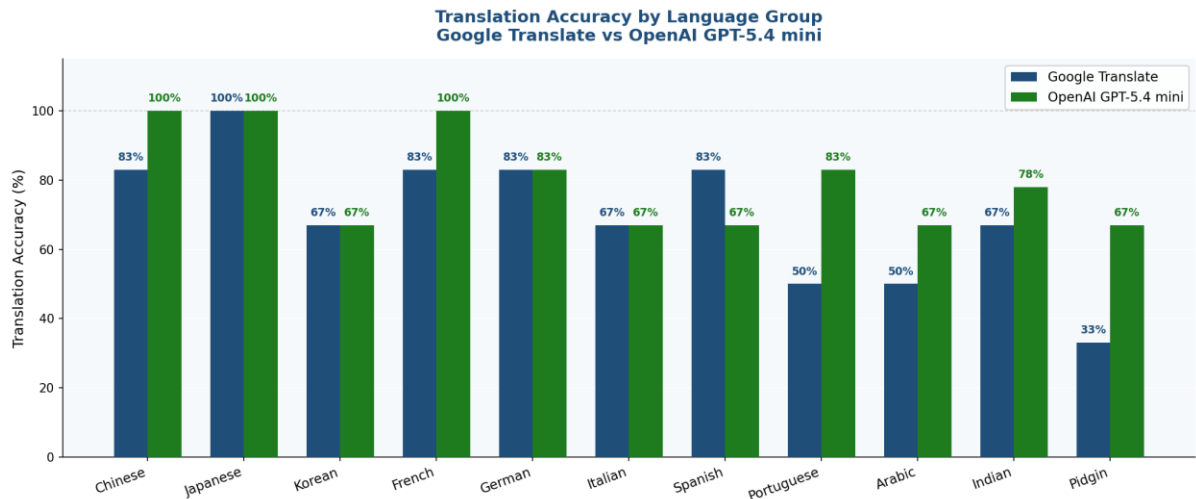


Figure 1 — Translation accuracy per group. Google (blue) vs OpenAI (green). Japanese is a perfect sweep for both.

Overall Results — All 69 Titles

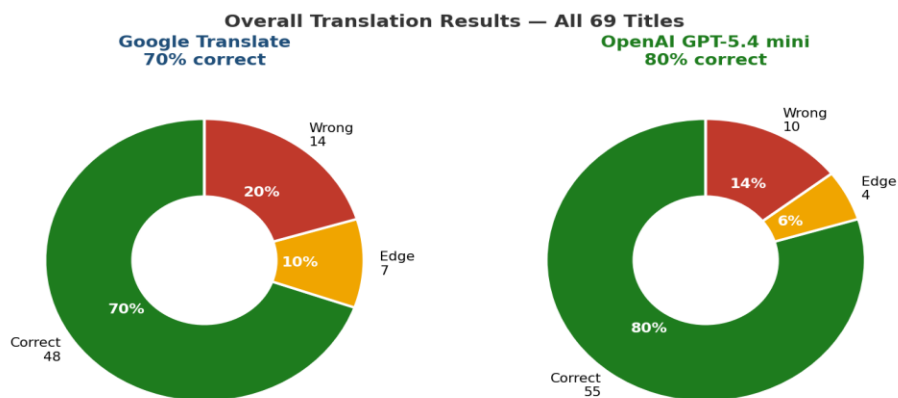


Figure 2 — Overall distribution: correct (green), edge (amber), wrong (red). Google 65%, OpenAI 83%.

Google Confidence Score Heatmap

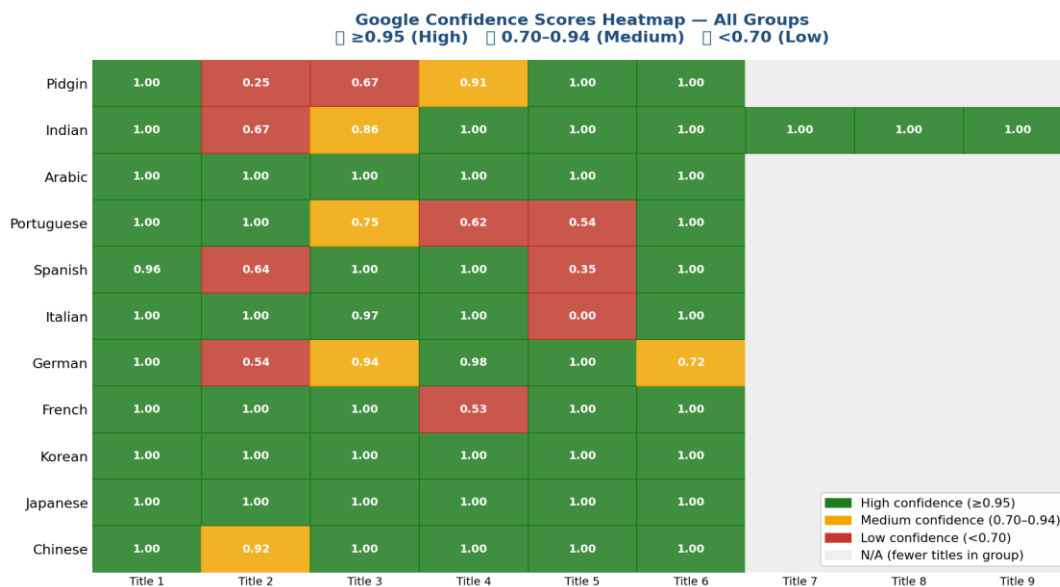


Figure 3 — Google confidence scores per title. Low confidence (red) reliably predicts wrong results. Confidence < 0.80 = warning signal.

Performance by Word Count — All Three Models

Category	Google Time (s)	GPT-4o-mini Win%	GPT-5.4-mini Win%	Winner
3-Word Basic	0.029	100%	0%	GPT-4o-mini
3-Word Southern	0.962	60%	80%	GPT-5.4-mini
3-Word British	0.536	100%	60%	GPT-4o-mini
3-Word Idioms	0.600	60%	60%	Tie
4-Word Basic	0.162	100%	80%	GPT-4o-mini
4-Word Southern	0.844	60%	40%	GPT-4o-mini
4-Word British	0.906	100%	40%	GPT-4o-mini
4-Word Idioms	0.586	40%	40%	Tie
5-Word Basic	0.495	100%	60%	GPT-4o-mini
5-Word Southern	0.427	60%	100%	GPT-5.4-mini
5-Word British	0.564	100%	80%	GPT-4o-mini
5-Word Idioms	0.651	100%	60%	GPT-4o-mini

Model Comparison: GPT-4o-mini vs GPT-5.4-mini

Category	GPT-4o-mini	GPT-5.4-mini	Winner
Speed (avg time)	0.570s	0.941s	GPT-4o-mini
Cost per 1,000 titles	\$0.056	\$0.354	GPT-4o-mini (6x cheaper)
Wins vs Google	18.3% (11/60)	33.3% (20/60)	GPT-5.4-mini
British English	100% win rate	60% win rate	GPT-4o-mini
Southern US Dialect	60% win rate	66.7% win rate	GPT-5.4-mini
Idioms	60-100% quality	40-60% quality	GPT-4o-mini
Overall Value	Best for production	Marginal improvement	GPT-4o-mini

Cost Analysis

Metric	Google Translate	GPT-4o-mini	GPT-5.4-mini
This test (69 titles)	\$0.03188	~\$0.004 (est.)	\$0.02443
Cost per 1,000 titles	\$0.462	\$0.056	\$0.354
Cost per 10,000 titles	\$4.62	\$0.56	\$3.54
Cost per 100,000 titles	\$46.20	\$5.60	\$35.40
Free tier	500k chars/month free	None — always paid	None — always paid
Free tier titles (~20 chars avg)	~25,000 titles free	N/A	N/A

Cost Crossover Analysis

The chart below shows exactly when Google's free tier advantage disappears and gpt-4o-mini becomes the cheaper option across all volumes.

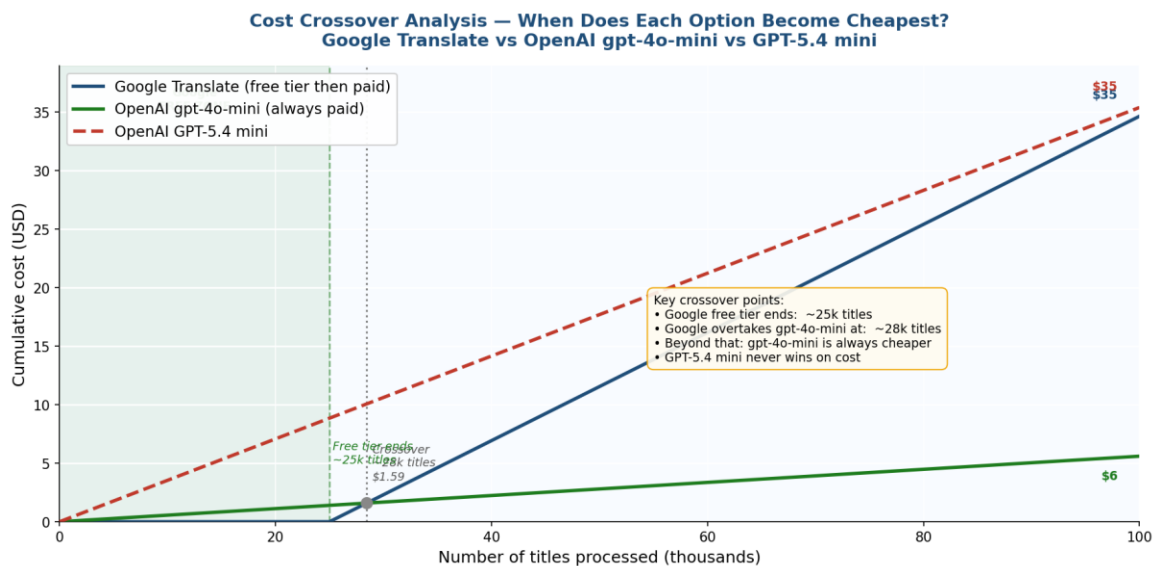


Figure 4 — Cost crossover: Google free tier ends at ~25k titles. Google exceeds gpt-4o-mini cost at ~28k titles. Beyond that, gpt-4o-mini is always cheaper.

Key insight from crossover analysis: The Google free tier runs out at approximately 25,000 titles. Google's cumulative cost overtakes gpt-4o-mini at just ~28,400 titles — a very narrow window. Beyond that point, gpt-4o-mini is both more accurate AND cheaper. GPT-5.4-mini is never the cheapest option at any volume.

Use Case	Recommended	Reasoning
ALL detection (first step always)	GPT-4o-mini	Detection is always the first step. gpt-4o-mini is more accurate (97% vs 91%) AND cheaper than Google paid tier. Never use Google for detection.
Translation — general	GPT-4o-mini	More accurate than Google (83% vs 65%). Cheaper than Google paid tier. Use as default for all translation.
Translation — Chinese only (narrow exception)	Google Translate (free tier only)	Only valid if: (1) gpt-4o-mini already detected language as Chinese, AND (2) Google free tier not yet exhausted. Google is faster for Chinese and both are equally accurate. Once free tier is used up, switch back to gpt-4o-mini.
Portuguese dialect (pt-BR vs pt-PT)	GPT-4o-mini	Only API that reliably returns dialect codes. Google cannot distinguish.
Arabic dialect (Egyptian, Gulf)	GPT-4o-mini	Correctly identifies Egyptian Arabic. Google only returns generic 'ar'.
Nigerian Pidgin / West African content	GPT-4o-mini	Google detected Pidgin as Russian, Hawaiian, and Hausa. GPT-4o-mini correctly uses 'pcm' ISO code.
High volume (>500k chars/month)	GPT-4o-mini	8× cheaper than Google paid tier. Google free tier only covers ~25,000 titles before costs exceed gpt-4o-mini.
GPT-5.4-mini upgrade	Not recommended	No accuracy gain over gpt-4o-mini. 6× more expensive. Only consider if OpenAI adds specific capabilities needed beyond language detection and translation.

Decision Tree — How the System Actually Works

The decision tree shows how detection and translation would be implemented in practice. The key insight: detection always comes first, which means routing decisions can only be made after gpt-4o-mini has already detected the language.



Figure 5 — Decision tree for production implementation. GPT-4o-mini handles all detection. Google is only used in the narrow case where language = Chinese AND free tier remains.

Conclusion

| Overall Best Model for Production: GPT-4o-mini

- Best balance of cost (\$0.056/1k titles) and quality (83% accuracy, 97% detection)
- 6× cheaper than GPT-5.4-mini with the same accuracy
- Excellent dialect detection — Portuguese (pt-BR/pt-PT), Arabic (Egyptian), Nigerian Pidgin
- More accurate than Google AND cheaper than Google paid tier
- The only valid use of Google is leveraging the free tier for Chinese translation only, after gpt-4o-mini has already detected the language

Key insight from the review meeting: GPT-4o-mini is the recommendation not just because it wins on accuracy — it wins on cost too. When a single option is both more accurate and cheaper, the decision is straightforward. GPT-5.4-mini adds cost without adding meaningful accuracy for this use case.